Gauravpreet Singh

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(Secret Clearance)

EDUCATION

University of Pittsburgh, Pittsburgh, PA | **GPA: 3.7** — Jan 2025 – May 2026
Master of Science, Electrical and Computer Engineering

Washington & Jefferson College, Washington, PA | **GPA: 3.6** — Aug 2020 – May 2024
Bachelors, Mathematics, Cum Laude

EXPERIENCE

Intern at Coherent*,* Pittsburgh, PA — Sept 2025 – Dec 2025

- Led the development of an AI-driven pipeline to automate the conversion of FAI (First Article Inspection) drawings into completed FAI documents.
- Engineered a multi-tool workflow leveraging Python, OpenCV, OCR, and LLMs to extract and analyze text from images
- Achieved over **80% accuracy** in auto-populating FAI sheets directly from engineering drawings, eliminating manual data entry and significantly reducing processing time.

Research Experiences for Undergraduates, Southern Methodist University — June 2023 – July 2023

- Applied mathematical modeling techniques to fit theoretical models to GISAXS (grazing-incidence small-angle X-ray scattering) experimental data.
- Implemented and validated model-fitting algorithms in **Python** to characterize nanoscale material structures.

Research Experiences for Undergraduates, Fairmont State University — May 2022 – Aug 2022

- Conducted theoretical research on **discrete Anger equations** and the **shift operator on quantum time scales**.
- Proved the discrete analogue of classical Anger function equations, contributing novel results to the field.
- Presented research findings at the **Joint Mathematics Meetings (JMM) 2023** in Boston, MA.

PUBLICATIONS

The Anger Difference Equation, *Fairmont State University* — May 2022 – Present

- Developed solutions for the non-homogeneous Bessel function, introducing the Anger difference equation.
- Formulated discrete analogues of Anger Hypergeometric series, Laplace functions, and recurrence relations.
- Research paper to appear in the Journal of Classical Functions, 2026.

KEY PROJECTS

Open Source Parallel Kalman Filter*, University of Pittsburgh* — Feb 2025 – May 2025

- Implemented a parallel Kalman Filter algorithm in an open-source project using OpenMP and MPI.
- Benchmarking showed up to 5× speedup over the serial implementation for processing large datasets.
- Optimized data handling for thousands of data points in parallel computing environments.

Examining Feebly Amicable Numbers*, Washington & Jefferson College* — Sept 2024 – May 2025

- Investigated properties of Amicable and Feebly Amicable numbers, extending known results.
- Researched density proofs of Amicable numbers following Paul Erdos' work.
- Successfully generalized some proofs to Feebly Amicable numbers, contributing to ongoing research in number theory.

SKILLS

Technical: Design for Manufacturing, Control Systems Modeling, Circuit design, Mathematical Modelling, AI/ML

Software: Data Analysis (Python, MATLAB, C++), CAD/CAM (SolidWorks, Fusion360, NX)

COURSEWORK

Courses: Robotics, Real Analysis, Abstract Algebra, Performance Analysis, Parallel Computer Architecture, Power Electronics, Optimization Methods, Machining, Electronics, Linear Control Theory, Nano-optics